

Exercise JA

- Q1. Let PQR be a triangle of area Δ with $a = 2$, $b = \frac{7}{2}$ and $c = \frac{5}{2}$, where a , b and c are the lengths of the sides of the triangle opposite to the angles at P, Q and R respectively.

Then $\frac{2 \sin P - \sin 2P}{2 \sin P + \sin 2P}$ equals.

- (A) $\frac{3}{4\Delta}$ (B) $\frac{45}{4\Delta}$ (C) $\left(\frac{3}{4\Delta}\right)^2$ (D) $\left(\frac{45}{4\Delta}\right)^2$

Ans. C

Sol.
$$\frac{2 \sin P - 2 \sin P \cos P}{2 \sin P + 2 \sin P \cos P} = \frac{(1 - \cos P)}{1 + \cos P}$$

$$= \tan^2 \frac{A}{2} = \frac{\Delta^2}{s^2(s-a)^2} = \frac{(s-b)(s-c)^2}{\Delta^2}$$

$$s = 4$$

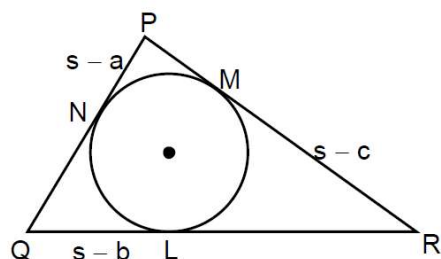
$$\left(\frac{(4 - \frac{7}{2})(4 - \frac{5}{2})^2}{\Delta} \right) = \left(\frac{3}{4\Delta} \right)^2$$

- Q2. In a triangle PQR, P is the largest angle and $\cos P = \frac{1}{3}$. Further the incircle of the triangle touches the sides PQ, QR and RP at N, L and M respectively, such that the lengths of PN, QL and RM are consecutive even integers. Then possible length (s) of the side(s) of the triangle is (are)

- (A) 16 (B) 18 (C) 24 (D) 22

Ans. D, B

Sol.



Let

$$s - a = 2k - 2, s - b = 2k, s - c = 2k + 2, k \in \mathbb{I}, k > 1$$

Adding we get,

$$s = 6k$$

$$\text{So, } a = 4k + 2, b = 4k, c = 4k - 2$$

$$\text{Now, } \cos P = \frac{1}{3}$$

$$\Rightarrow \frac{b^2 + c^2 - a^2}{2bc} = \frac{1}{3} \Rightarrow 3 [(4k)^2 + (4k - 2)^2 -] = 2 \times 4k (4k - 2)$$

$$\Rightarrow 3 [16k^2 - 4(k) \times 2] = 8k (4k - 2)$$

$$\Rightarrow 48k^2 - 96k = 32k^2 - 16k$$

$$\Rightarrow 16k^2 = 80k \Rightarrow k = 5$$

So, sides are 22, 20, 18

Q3. In a triangle the sum of two sides is x and the product of the same two sides is y . If $x^2 - c^2 = y$, where c is the third side of the triangle, then the ratio of the in-radius to the circum-radius of the triangle is

(A) $\frac{3y}{2x(x+c)}$ (B) $\frac{3y}{2c(x+c)}$ (C) $\frac{3y}{4x(x+c)}$ (D) $\frac{3y}{4c(x+c)}$

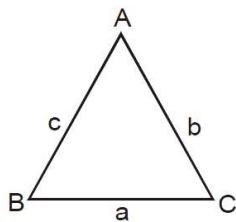
Ans. B

Sol. $a + b = x$

$$ab = y$$

$$x^2 - c^2 = y$$

$$(a + b)^2 = ab \quad a^2 + b^2 - c^2 = -ab$$



$$\frac{a^2 + b^2 - c^2}{2ab} = \frac{7}{2}$$

$$\cos C = \frac{-1}{2}$$

$$C = \frac{2\pi}{3}$$

$$\frac{r}{R} = \frac{\Delta \times 4\Delta}{s \times abc} = \frac{4 \times \frac{1}{4} a^2 b^2 \sin^2 C}{(a + b + c) abc} = \frac{3ab}{4c(x + c)} = \frac{3y}{4c(x + c)}$$

Q4. In a triangle XYZ, let x, y, z be the lengths of sides opposite to the angles X, Y, Z , respectively, and $2s = x + y + z$. If $\frac{s-x}{4} = \frac{s-y}{3} = \frac{s-z}{2}$ and area of incircle of the triangle XYZ is $\frac{8\pi}{3}$, then

(A) area of the triangle XYZ is $6\sqrt{6}$

(B) the radius of circumcircle of the triangle XYZ is $\frac{35}{6}\sqrt{6}$

(C) $\sin \frac{X}{2} \sin \frac{Y}{2} \sin \frac{Z}{2} = \frac{4}{35}$ (D) $\sin^2 \left(\frac{x+y}{2} \right) = \frac{3}{5}$

Ans. D, C, A

Sol. $\frac{s-x}{4} = \frac{s-y}{3} = \frac{s-z}{9}$

$$\Rightarrow \Delta = \frac{2\sqrt{6}}{27}s^2 \text{ and inradius } r = \frac{2\sqrt{6}}{27}s$$

$$\Rightarrow s = 9, x = 5, y = 6, z = 7$$

$$\Rightarrow \sin \frac{X}{2} \sin \frac{Y}{2} \sin \frac{Z}{2} = \frac{4}{35}$$

$$= (1 + \cos Z) = \frac{1}{2} = \left(1 + \frac{1}{5}\right) = \frac{3}{5}$$

Q5. The sides of a right angled triangle are in arithmetic progression. If the triangle has area 24, then what is the length of its smallest side?

Ans. 6

Sol. Sides will be $a-d, a, a+d$

$$(a-d)^2 + a^2 = (a+d)^2$$

$$d = +\frac{a}{4}$$

So, sides will be $\frac{3a}{4}, a, \frac{5a}{4}$

$$\frac{1}{2} \times \frac{3a}{4} \times a = 4$$

$$a^2 = (8) \Rightarrow a = 8$$

Sides will be 6, 8, 10

Q6. In a triangle PQR, let $\angle PQR = 30^\circ$ and the sides PQ and QR have lengths $10\sqrt{3}$ and 10 , respectively. Then, which of the following statement(s) is (are) TRUE ?

(A) $\angle QPR = 45^\circ$ (B) The area of the triangle PQR is $25\sqrt{3}$ and $\angle QRP = 120^\circ$

(C) The radius of the incircle of the triangle PQR is $10\sqrt{3}-15$

(D) The area of the circumcircle of the triangle PQR is 100π

Ans. B, C, D

Sol. $\cos Q = \frac{100 + 300 - (PR)^2}{2 \cdot 10 \cdot 10\sqrt{3}} \Rightarrow \frac{\sqrt{3}}{2} = \frac{100 + 300 - (PR)^2}{2 \cdot 10 \cdot 10\sqrt{3}}$

$$300 = 400 - (PR)^2 \Rightarrow PR = 10$$

$$\Delta = \frac{1}{2}(PQ)(QR)\sin Q = \frac{1}{2} 10 \cdot 10\sqrt{3} \times \frac{1}{2} = 25\sqrt{3}$$

$$r = \frac{\Delta}{s} = \frac{25\sqrt{3} \times 2}{(20 + 10\sqrt{3})} = \frac{50\sqrt{3}}{20 + 10\sqrt{3}} = \frac{5\sqrt{3}}{2 + \sqrt{3}} \times \frac{2 - \sqrt{3}}{2 - \sqrt{3}} = 5(2\sqrt{3} - 3) = 10\sqrt{3} - 15$$

$$\text{by sin rule } \frac{10\sqrt{3}}{\sin R} = \frac{10}{\sin Q} \Rightarrow \angle R = 30$$

$$2(\text{circumradius}) = \frac{PR}{\sin Q} = \frac{10}{1/2} \Rightarrow \text{circumradius} = 10$$

$$\text{Hence area of circumcircle} = \pi R^2 = 100\pi$$

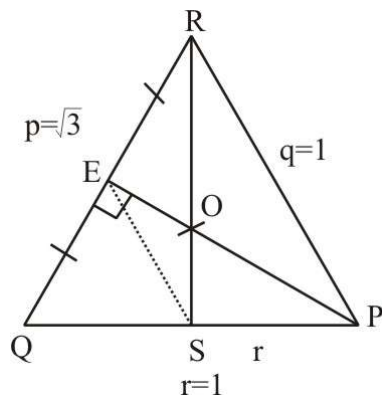
Q7. In a non right angled triangle ΔPQR let p, q, r denote the length of the sides opposite to the angles at P, Q, R respectively. The median from R meets the side PQ at S , the perpendicular from P meets the side QR at E , and RS and PE intersect at O . If $p = \sqrt{3}$, $q = 1$ and the radius of the circumcircle of the ΔPQR equals 1, then which of the following options is/are correct

$$(A) \text{ Length of } OE = \frac{1}{6} \quad (B) \text{ Length of } RS = \frac{\sqrt{7}}{2}$$

$$(C) \text{ Radius of incircle of } \Delta PQR = \frac{\sqrt{3}}{2}(2 - \sqrt{3}) \quad (D) \text{ Area of } \Delta SOE = \frac{\sqrt{3}}{12}$$

Ans. C, B, A

Sol.



$$\frac{p}{\sin P} = \frac{q}{\sin Q} = \frac{r}{\sin R} = 2R$$

$$\frac{\sqrt{3}}{2} = \sin P, = \frac{1}{2} \sin Q$$

$$P = \frac{\pi}{3}, Q = \frac{\pi}{6} \text{ then } R = \frac{\pi}{2} \text{ (not possible)}$$

$$\Delta = \frac{1}{2} 1 \times 1 \sin \frac{2\pi}{2} = \frac{1}{2} \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{4}$$

$$s = \frac{1 + 1 + \sqrt{3}}{2} = \frac{2 + \sqrt{3}}{2}$$

$$r = \frac{\sqrt{3}}{4} \times \frac{2}{(2 + \sqrt{3})} = \frac{\sqrt{3}}{4} \times \frac{1}{(2 + \sqrt{3})}$$

$$r = \frac{\sqrt{3}}{2} (2 - \sqrt{3})$$

$$RS = \frac{1}{2} \sqrt{2(3) + 2(1) - 1} = \frac{\sqrt{7}}{2}$$

$$OE = PE \Rightarrow OE = \frac{1}{3} \cdot \frac{1}{2} \sqrt{2(1) + 2(1) - 3} = \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$$

$$\text{Area of } \triangle SOE = \frac{1}{12} (\text{area of } \triangle PQR) = \frac{\sqrt{3}}{48}$$

Q8. Let x, y and z be positive real numbers. Suppose x, y and z are the length of the sides of a triangle opposite to its angles X, Y and Z , respectively. If $\tan \frac{X}{2} + \tan \frac{Z}{2} = \frac{2y}{x+y+z}$ then which of the following statements is/are TRUE?

(A) $2Y = X + Z$ (B) $Y = X + Z$ (C) $\tan \frac{X}{2} = \frac{x}{y+z}$ (D) $x^2 + z^2 - y^2 = xz$

Ans. B, C

Sol. $\tan\left(\frac{X}{2}\right) + \tan\left(\frac{Z}{2}\right) = \frac{2y}{x+y+z}$

$$\frac{\Delta}{s(s-x)} + \frac{\Delta}{s(s-z)} = \frac{2y}{2s}$$

$$\Delta \left(\frac{1}{s-x} + \frac{1}{s-z} \right) = \frac{2y}{2s}$$

$$\frac{\Delta(s-z) + (s-x)}{(s-x)(s-z)} = y$$

$$\frac{\Delta y}{(s-x)(s-z)} = y$$

$$\cot\left(\frac{Y}{2}\right) = 1$$

$$\frac{Y}{2} = \frac{\pi}{4} \Rightarrow Y = \frac{\pi}{2}$$

$$\Rightarrow Y = X + Z = \frac{\pi}{2} \quad \Rightarrow \text{A is false / B is true}$$

(c) $\tan\left(\frac{X}{2}\right) = \frac{x}{y+z}$

$$\frac{x}{\sin(X)} = \frac{y}{\sin(90^\circ)} = \frac{z}{\sin(90^\circ - x)}$$

$$RHS = \frac{y \sin(X)}{y + y \cos X} = \frac{\sin(X)}{1 + \cos(X)}$$

$$= \tan\left(\frac{X}{2}\right) = LHS$$

$$(b) x^2 + z^2 - y^2 = xz$$

$$\frac{x^2 + z^2 - y^2}{2zx} = \frac{1}{2}$$

$$\cos Y = \frac{1}{2} \Rightarrow Y = \frac{\pi}{3} \text{ D is false}$$

Q9. Consider a triangle PQR having sides of lengths p, q and r opposite to the angles P, Q and R, respectively. Then which of the following statement is (are) TRUE ?

$$(A) \cos P \geq 1 - \frac{p^2}{2qr} \quad (B) \cos R \geq \left(\frac{q-r}{p+q} \right) \cos P + \left(\frac{p-r}{p+q} \right) \cos Q$$

$$(C) \frac{q+r}{p} < 2 \frac{\sqrt{\sin Q \sin R}}{\sin R} \quad (D) \text{ If } p < q \text{ and } p < r, \text{ then } \cos Q > \frac{p}{r} \text{ and } \cos R > \frac{p}{q}$$

Ans. A, B

Sol.

$$(A) \cos P = \frac{q^2 + r^2 - p^2}{2qr}$$

$$\cos P = \frac{1}{2} \left(\frac{q}{r} + \frac{r}{q} \right) - \frac{p^2}{2qr} \quad \left(\frac{q}{r} + \frac{r}{q} \geq 2 \right)$$

$$\cos P \geq 1 - \frac{p^2}{2qr}$$

$$(B) \text{ Let } \cos R \geq \left(\frac{q-r}{p+q} \right) \cos P + \left(\frac{p-r}{p+q} \right) \cos Q$$

$$(p+q) \cos R \geq (q-r) \cos P + (p-r) \cos Q$$

$$(p \cos R + r \cos P) + (q \cos R + r \cos Q) \geq (q \cos P + p \cos Q)$$

$$q + p \geq r \text{ (Triangle inequality)}$$

$$(C) \text{ Let } \frac{q+r}{p} < 2 \frac{\sqrt{\sin q \sin R}}{\sin P}$$

$$\Rightarrow \frac{\sin Q + \sin R}{2} < \sqrt{\sin Q \sin R}$$

$$AM < GM \text{ (Not possible)}$$

$$(D) \text{ Let } \cos > \frac{p}{r} \text{ and } \cos R > \frac{p}{q}$$

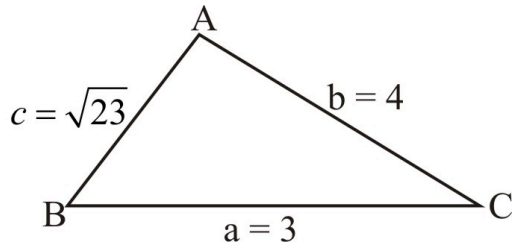
$$\frac{r^2 + p^2 - q^2}{2rp} \text{ and } \frac{p^2 + q^2 - r^2}{2pq} > \frac{p}{q}$$

$$\Rightarrow r^2 > p^2 + q^2 \text{ and } q^2 > p^2 + r^2 \text{ [Not possible]}$$

Q10. In a triangle ABC, let $AB = \sqrt{23}$, $BC = 3$ and $CA = 4$. Then the value of $\frac{\cot A + \cot C}{\cot B}$ is ____.

Ans. 2

Sol.



According to sin rule & cosine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = \frac{1}{\lambda} \text{ (say)}$$

$$\Rightarrow \sin A = a\lambda$$

$$\Rightarrow \sin B = b\lambda$$

$$\Rightarrow \sin C = c\lambda$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}, \cos B = \frac{a^2 + c^2 - b^2}{2ac}, \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\text{Now, } \frac{\cot A + \cot C}{\cot B}$$

$$= \frac{\frac{\cos A}{\sin A} + \frac{\cos C}{\sin C}}{\frac{\cos B}{\sin B}} = \frac{\frac{b^2 + c^2 - a^2}{2bc(a\lambda)} + \frac{a^2 + b^2 - c^2}{2ab(c\lambda)}}{\frac{a^2 + c^2 - b^2}{2ac(b\lambda)}}$$

$$= \frac{b^2 + c^2 - a^2 + a^2 + b^2 - c^2}{a^2 + c^2 - b^2} = \frac{2b^2}{a^2 + c^2 - b^2}$$

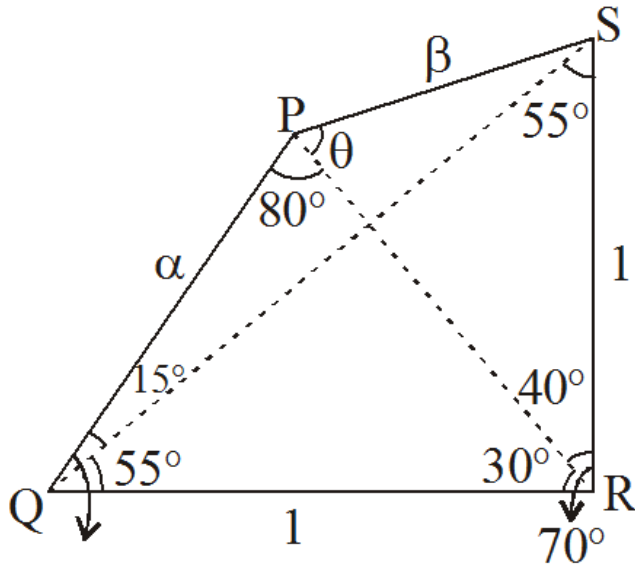
$$= \frac{2 \times 16}{9 + 23 - 16} = \frac{32}{16} = 2$$

Q11. Let PQRS be a quadrilateral in plane, where $QR = 1$, $\angle PQR = \angle QRS = 70^\circ$, $\angle PQS = 15^\circ$ and $\angle PRS = 40^\circ$. If $\angle RPS = \theta$, $PQ = \alpha$ and $PS = \beta$, then interval(s) that contain(s) the value of $4\alpha\beta \sin \theta^\circ$ is /are :

(A) $(0, \sqrt{2})$ (B) $(1, 2)$ (C) $(\sqrt{2}, 3)$ (D) $(2\sqrt{2}, 3\sqrt{2})$

Ans. A, B

Sol. Figure as shown, can be drawn on the basis of given data.



Applying sine rule in ΔPQR ,

$$\frac{\alpha}{\sin 30^\circ} = \frac{1}{\sin 80^\circ}$$

$$\Rightarrow \alpha = \frac{1}{2 \sin 80^\circ} \quad \text{_____ (i)}$$

Applying sine rule in ΔPRS ,

$$\frac{\beta}{\sin 40^\circ} = \frac{1}{\sin \theta}$$

$$\Rightarrow \beta \sin \theta = \sin 40^\circ \quad \text{_____ (ii)}$$

From (i) & (ii),

$$4\alpha\beta \sin \theta = 4 \cdot \frac{1}{2 \sin 80^\circ} \cdot \sin 40^\circ = \frac{1}{\cos 40^\circ}$$

$$\text{Using } \cos 0 = 1, \cos 30^\circ = \frac{\sqrt{3}}{2}, \cos 45^\circ = \frac{1}{\sqrt{2}} \text{ and } \cos 60^\circ = \frac{1}{2}$$

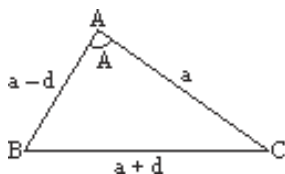
Only Options (A) and (B) are correct

- Q12. Consider an obtuse angled triangle ABC in which the difference between the largest and the smallest angle is $\frac{\pi}{2}$ and whose sides are in arithmetic progression. Suppose that the vertices of this triangle lie on a circle of radius 1.

Let a be the area of the triangle ABC. Then the value of $(64a)^2$ is

Ans. 1008

Sol.



Let sides be $a - d, a, a + d$

$$A - C = \frac{\pi}{2}$$

$$R = 1$$

Now

$$\frac{a+d}{\sin A} = \frac{a}{\sin B} = \frac{a-d}{\sin C} = 2$$

$$A = \frac{\pi}{2} + C$$

$$\sin A = \sin\left(\frac{\pi}{2} + C\right)$$

$$\sin A = \cos C.$$

$$\frac{a+d}{2} = \sqrt{1 - \sin^2 C}$$

$$\left(\frac{a+d}{2}\right)^2 = 1 - \left(\frac{a-d}{2}\right)^2$$

$$\frac{2(a^2 + d^2)}{4} = 1$$

$$a^2 + d^2 = 2 \quad \text{_____ (1)}$$

$$\text{Now } \cos B = \frac{(a-d)^2 + (a+d)^2 - a^2}{2(a^2 - d^2)}$$

$$\sqrt{1 - \sin^2 B} = \frac{2(a^2 + d^2) - a^2}{2(a^2 - d^2)}$$

$$\sqrt{1 - \frac{a^2}{4}} = \frac{4 - a^2}{2(a^2 - d^2)} \quad (a^2 + d^2 = 2)$$

$$(a^2 - d^2)^2 = 4 - a^2 \quad \text{_____ (2)}$$

From (1) and (2)

$$a^2 = \frac{7}{4}, d^2 = \frac{1}{4}$$

Area of triangle

$$\Delta = \frac{a(a^2 - d^2)}{4}$$

$$\alpha = \frac{\sqrt{7}}{2} \times \frac{6}{4 \times 4}$$

$$(64 \alpha)^2 = 1008$$

- Q13. Consider an obtuse angled triangle ABC in which the difference between the largest and the smallest angle is $\frac{\pi}{2}$ and whose sides are in arithmetic progression. Suppose that the vertices of this triangle lie on a circle of radius 1.

Then the inradius of the triangle ABC is

Ans. 0.25

Sol.
$$r = \frac{\Delta}{S} = \frac{\frac{\sqrt{7}}{2} \times \frac{6}{16}}{\frac{3}{2} \times \frac{\sqrt{7}}{2}} = \frac{4}{16} = \frac{1}{4} = 0.25$$